SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

C01_AT2 — DATA INTERPRETATION EXERCISE

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO1 –Apply (BL3)	**Assessment:**	Assessment Tool 2 – Data Interpretation
Weightage:	20% of CIA	**Total Marks:**	20(2 Questions × 20 Marks)
CO1 Statement:	*Understand the fundamental concepts, principles, and techniques of data visualization.(BL2)*		
Student Name:	K.Amrutha Sandhya	**Reg. No.:**	192524270
Section / Batch:	AIDS	**Date:**	28-07-2026

Code of Conduct

I K.Amrutha Sandhya(192524270) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Amrutha

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.

Annexure A – DATA INTERPRETATION

Rubrics for Calculation ASSESSMENT RUBRIC

AT2 — Data Interpretation Exercise

CO1 · BT3: Apply ·

This rubric is used to assess student responses to the AT2 Data Interpretation Exercise problems, in which students interpret a described data visualization (data types, aesthetic mappings, scales, coordinate systems, and color scales) and answer accompanying sub-questions.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

Understanding of Concepts	**25%**	Demonstrates thorough, accurate understanding of data types (nominal, ordinal, numerical), aesthetic mappings, scales, coordinate systems, and color scales as they apply to the given visualization scenario.	Good understanding of the relevant visualization concepts, with only minor gaps in identifying data types, aesthetic mappings, or scale choices.	Basic understanding of visualization concepts, but with some misconceptions about data types, aesthetics, coordinate systems, or color scales.	Poor understanding of core visualization concepts; major misconceptions about data types, aesthetics, coordinate systems, or color scales.
Explanation	**25%**	Provides detailed, well-reasoned explanations for each sub question, using specific details from the scenario and correct visualization terminology.	Provides clear explanations with adequate justification, referencing the scenario appropriately for most sub questions.	Explanations are limited or superficial; lacks depth or fails to consistently connect reasoning to the given scenario.	Explanations are poor, unclear, or missing; answers are unsupported assertions with no reasoning shown.
Application	**20%**	Effectively applies visualization concepts to interpret the specific scenario, drawing accurate, well-reasoned conclusions from the described chart or data pattern.	Applies concepts to interpret the scenario correctly, with only minor errors in reasoning or in the conclusions drawn.	Limited application of concepts to the scenario; conclusions are weakly supported or only partially correct.	Fails to apply concepts to the scenario, or the conclusions drawn are incorrect or irrelevant to the data shown.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

Organization	**15%**	Answers are well structured, addressing each sub-question (a, b, c) in a clear, logical sequence with coherent flow throughout.	Answers are organized and address each sub-question, with only minor issues in flow or clarity.	Basic structure; sub-questions are addressed but answers lack clarity or a logical sequence.	Poorly organized; answers are incoherent, and sub-questions are left unaddressed or answered out of order.
Accuracy	**10%**	All parts of the answer — data type identification, aesthetic/scale justification, and final interpretation — are completely accurate.	Mostly accurate, with only minor omissions or a small error in one part of the answer.	Partially correct; some key points (e.g., data type, aesthetic justification, or interpretation) are missing or incorrect.	Answers are largely incorrect, with major gaps in identifying data types, aesthetics, scales, or drawing conclusions.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	**Course Coordinator**	**Head of Department**
Signature: _______________	Signature: _______________	Signature: _______________
Date: ___________________	Date: ___________________	Date: ___________________

AT2 — Data Interpretation Exercise

Problem 4: Bar Chart Aesthetic Mapping & Interpretation Risk

Scenario Description:

A bar chart shows the number of products sold in four categories (Electronics, Clothing, Books, Toys), with bar height mapped to unit count and bar width kept constant.

Part (a) Which aesthetic mapping encodes the primary data value in

this chart?

Position along a common scale (specifically, vertical position or bar height) encodes the primary quantitative data value (unit count).

According to perceptual hierarchy research (Cleveland & McGill), position along a common scale is the most accurately perceived visual aesthetic mapping. Mapping the numerical variable 'unit count' to the y-axis position allows viewers to decode exact quantitative values and compare magnitude differences across discrete nominal categories with maximum precision.

The visual mark used is a rectangle, where horizontal length (bar width) is intentionally held constant as a non-data-driven aesthetic, isolating vertical spatial position (bar height) as the sole visual channel for the primary data variable.

Part (b) Would size (bar width) have been a valid additional aesthetic to encode a second variable, such as profit margin? Explain.

No. Utilizing size (bar width) to encode a second variable is strongly discouraged in sound data visualization practice.

When both height (units sold) and width (profit margin) are varied simultaneously, human visual perception automatically integrates the two dimensions into a single Gestalt property: total 2D area (Height × Width). Viewers struggle to separate individual 1D attributes (width vs. height) when presented in a 2D mark, leading to severe visual distortion:

• Area Disproportion (Non-Linear Perceptual Scaling): Doubling the width of a tall bar quadruples its total 2D visual surface area, causing small variations in profit margin to appear visually overwhelming and disproportionately exaggerated relative to adjacent categories.

• Conflation of Independent Variables: Mathematically, Area = Units Sold × Profit Margin = Total Profit. While total profit is a meaningful metric, viewers cannot accurately decode or isolate the underlying profit margin percentage directly from bar width, creating ambiguity between the variable being explicitly encoded and the perceived area.

Part (c) If 'Electronics' has the tallest bar but the smallest profit margin (not shown), what interpretation risk does this chart create for a viewer?

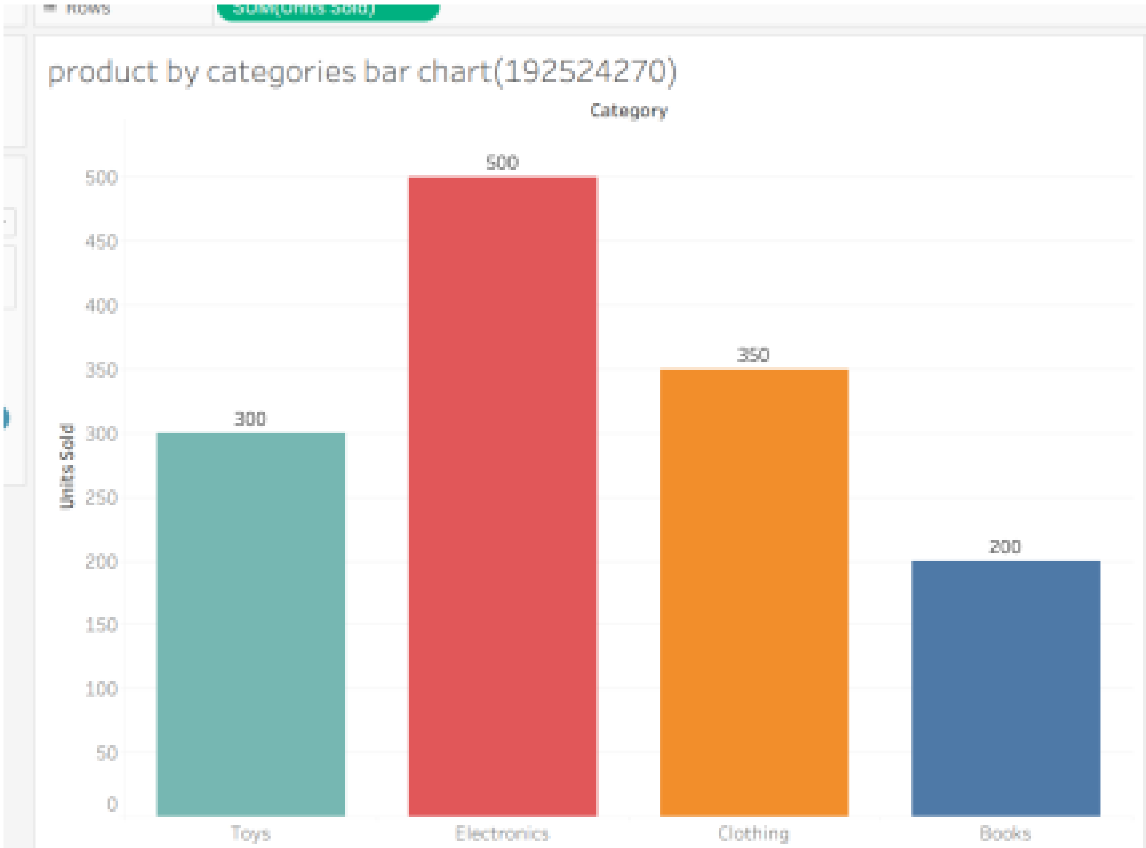
The primary interpretation risk is Salience Bias (also known as Visual Dominance Bias or False Equivalence).

Because bar height and overall visual surface area are the most visually prominent features of the chart, viewers naturally apply an intuitive heuristic that 'tallest visual mark = top-performing or most valuable category.'

Without profit margin data displayed, a viewer or business decision-maker is at risk of drawing misleading conclusions:

• Misattributing Revenue/Profitability to Volume: High unit sales (volume) do not necessarily correlate with high net financial return. Electronics may generate significant operational volume while contributing lower net profit than a smaller volume category with healthy profit margins (e.g., Books or Clothing).

• Strategic Misallocation of Capital: Relying strictly on this incomplete visualization could lead executives to over-invest marketing, floor space, or inventory budget into high-volume/low-margin items, misaligning business strategy with overall net profitability.



Problem 14: GPS Spatial Coordinates & Map Projections

Scenario Description:

A GPS tracking app plots a delivery driver's location every 10 minutes over a single day using standard Cartesian (X, Y) coordinates on a city map background.

Part (a) What do the X and Y axes represent in this context?

• X-axis: Represents Longitude (or Easting / horizontal spatial coordinate on the planar map grid), indicating East/West displacement.

• Y-axis: Represents Latitude (or Northing / vertical spatial coordinate on the planar map grid), indicating North/South displacement.

On a 2D planar map projection (such as Web Mercator or UTM), these axes transform 3D spherical geographic coordinates into a standard Cartesian orthogonal plane overlaying the city's street network.

Part (b) If the driver returns to the same starting point at the end of the

day, how would that be reflected in the coordinate data?

In the raw coordinate dataset, the final data point record at time step t_final will contain exact numerical identity with the initial data point record at time step t_0, such that (X_final, Y_final) = (X_0, Y_0).

In the visual map layer, the sequential spatial path (geom_path) connecting the time ordered coordinates will form a completely closed geometric polygon/loop.

From a spatial analytical perspective, the net displacement vector equals zero ($\Delta X = 0$, $\Delta Y = 0$), whereas the cumulative distance along the path represents the non-zero positive route mileage.

Part (c) Would polar coordinates be a more or less natural choice for this specific scenario? Justify your answer.

Polar coordinates would be significantly LESS natural for this scenario.

• Mismatch with Built Environment Infrastructure: Modern urban street grids, block networks, and buildings are constructed along orthogonal, Cartesian spatial systems (North/South and East/West directions). Vehicle travel occurs along linear orthogonal vectors, matching Cartesian coordinates naturally.

• Distortion of Geometries and Distances: Polar coordinates define locations via radial distance (r) and angular displacement (θ) relative to a single central origin point. Converting planar city street networks to polar coordinates severely warps straight roads into curved radial arcs and distorts relative distances depending on proximity to the central origin.

• Absence of a Single Rotational Hub: Delivery routes serve distributed network points across a city rather than orbiting around a central hub. Imposing a polar system introduces unnecessary computational complexity and yields an unintuitive map projection.

Scenario Description:

A bar chart shows the number of products sold in four categories (Electronics, Clothing, Books, Toys), with bar height mapped to unit count and bar width kept constant.

Scenario Description:

A GPS tracking app plots a delivery driver's location every 10 minutes over a single day using standard Cartesian (X, Y) coordinates on a city map background.

gps map (192524270)